

Amber Li

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EDUCATION

Carnegie Mellon University

Pittsburgh, PA

Doctor of Philosophy in Robotics (GPA: 5.0/5.0)

May 2029

Relevant coursework: Robot Learning, Generative AI, Planning and Decision-making in Robotics, Integrated Planning and Learning, Learning for 3D Vision

Massachusetts Institute of Technology

Cambridge, MA

Master of Engineering in Computer Science: concentration in Artificial Intelligence (GPA: 5.0/5.0)

February 2023

Bachelor of Science in Computer Science, minor in Mathematics (GPA: 4.9/5.0)

May 2022

SKILLS

Programming: Python, Java, C

Libraries & Tools: PyTorch, NumPy, PyTorch3D, PyBullet, ManiSkill, pandas, Git, LaTeX

Languages: English (native), Mandarin Chinese (conversant)

PUBLICATIONS

Kallol Saha*, **Amber Li***, Ángela Rodríguez-Izquierdo*, Lifan Yu, Ben Eisner, Maxim Likhachev, David Held. Planning from Point Clouds over Continuous Actions for Multi-object Rearrangement. Conference on Robot Learning, 2025. **Oral presentation (Top 5.7%)**.

Amber Li, Tom Silver. Embodied Active Learning of Relational State Abstractions for Bilevel Planning. Conference on Lifelong Learning Agents, 2023. **Oral presentation (Top 12 papers)**.

RESEARCH EXPERIENCE

Robots Perceiving and Doing Lab, Search-Based Planning Lab

Pittsburgh, PA

Graduate Student Researcher (Advisors: David Held, Maxim Likhachev)

September 2024 – Present

- Developing methods that combine foundation-model priors with online planning and interaction to enable long-horizon robotic manipulation, adaptation, and physical reasoning
- Developed a self-supervised framework that adapts an action-conditioned visual world model to unseen tool-use tasks using VLM-proposed plans and their executed outcomes
- Built a hybrid learning-and-planning approach that uses learned models as priors to guide search in high-dimensional continuous action spaces, enabling multi-object rearrangement directly from point clouds

Basis Research Institute

New York, NY

Research Intern (Advisors: Zenna Tavares, Kevin Ellis, Tom Silver)

May 2026 – August 2026

- Built a real-world robotic agent that actively learns abstract world models and uses them for planning and reasoning
- Developed the perception and learning pipeline for discovering relational abstractions and inferring physical parameters, enabling the agent to manipulate dominoes and complete planning tasks
- Built a Franka Panda research workbench using the DROID hardware stack

Learning and Intelligent Systems Group

Cambridge, MA

Graduate Student Researcher (Advisor: Leslie Kaelbling)

September 2021 – January 2023

- Contributed to a bilevel task and motion planning framework by proposing an embodied active learning algorithm for learning neuro-symbolic relational state abstractions
- Implemented the algorithm and evaluated it in three robotic planning environments, demonstrating improved sample efficiency and outperforming all baselines

Computer-Aided Programming Group

Cambridge, MA

Undergraduate Student Researcher (Advisor: Armando Solar-Lezama)

December 2020 – May 2021

- Investigated how governance structures could be optimized for specified societal objectives using reinforcement learning and program synthesis

Marin Soljačić Group

Cambridge, MA

Undergraduate Student Researcher (Advisor: Marin Soljačić)

September 2020 – May 2021

- Constructed a graph neural network model that can be trained end-to-end while enabling extraction of interpretable equations
- Validated this approach on data describing particle interaction forces
- Demonstrated the advantages of the architecture for symbolic regression over simpler approaches

PROFESSIONAL EXPERIENCE

Two Sigma Investments

New York, NY

Software Engineer

February 2023 – June 2024

- Developed data pipeline infrastructure, monitoring dashboards, and profiling tools to ensure data availability and quality for modeling partners
- Migrated legacy pipelines and performed inventory of 200+ pipelines to reduce team-wide operational support burden

TEACHING EXPERIENCE

Carnegie Mellon University

Pittsburgh, PA

Planning and Decision-making in Robotics | *Teaching Assistant*

Fall 2026

Massachusetts Institute of Technology

Cambridge, MA

Introduction to Algorithms | *Graduate Teaching Assistant*

Spring 2023, Fall 2023

HAUSCR Summit for Young Leaders in China

Virtual

How to Give Presentations | *Core Lecturer*

August 2020

Massachusetts Institute of Technology

Cambridge, MA

Introduction to Algorithms | *Grader, Lab Assistant*

Spring 2020, Fall 2020

Fundamentals of Programming | *Lab Assistant*

Fall 2019

LEADERSHIP

MIT Women in EECS | *Mentorship Committee Member, Co-President*

September 2019 – May 2022

- Elected co-president, leading community, mentorship, and industry-engagement initiatives
- Led biweekly executive meetings, collaborated with EECS admin, and drove the creation of the organization's first corporate partnership program
- Organized mentor-mentee matches and planned monthly events for the mentorship program with 100+ members

Society of Women Engineers (SWE) – MIT Chapter | *Technology Chair*

February 2019 – December 2020

- Organized and taught technology workshops to SWE members and the MIT community
- Mentored middle school girls in web development using HTML/CSS and JavaScript

ACTIVITIES & AWARDS

CMU Manipulation Discussion Group | *Organizer*

September 2025 – Present

Learning Effective Abstractions for Planning @ CoRL 2026 | *Workshop Co-organizer*

November 2026

Conference on Robot Learning | *Reviewer*

2025, 2026

CMU Robotic Vision Fellowship | *Awardee*

October 2025

CMU All-University Orchestra | *First Violinist*

September 2024 – Present

HKN (Eta Kappa Nu, EECS Honor Society) – MIT Chapter | *Member, Tutor*

May 2020 – May 2022

MIT Chamber Music Society | *Violinist, Chamber Music Performance Awardee*

September 2019 – December 2022